Measuring Housing Displacement in Los Angeles County: The Role of Rent Burden, Homeownership Disparities, and the COVID-19 Pandemic

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Abstract

The COVID-19 pandemic had a big impact on the U.S. economy, making housing affordability problems worse and widening gaps in homeownership and rent burden. This study looks at how race and income levels affected housing in Los Angeles County before and after the pandemic. We used data from the five-year American Community Survey (ACS), where the 2014–2018 period represents pre-COVID and the 2018–2022 period represents post-COVID. Our focus was on changes in rent burden, homeownership rates, and median income across different areas.

To study these changes, we used Ordinary Least Squares (OLS) regression models. We grouped income into low, middle, and high categories based on median income percentiles and assigned a dominant racial group to each area for analysis.

We found that the rent burden stayed about the same, even though COVID-19 caused major economic challenges. This suggests that short-term policies like eviction bans and rental assistance may have helped keep rent burden from rising. However, while homeownership rates appeared to rise overall, our analysis suggests that this trend was not significantly driven by high-income households in the post-COVID period. Middle- and low-income households also did not experience meaningful gains, reinforcing existing disparities.

Our study highlights ongoing housing inequalities and provides useful data to help guide policy decisions. We recommend expanding affordable housing programs, increasing financial support for first-time homebuyers, and offering long-term rental assistance for low-income and minority communities. While COVID-19 did not cause a big rise in rent burden, it worsened existing differences in homeownership and income, showing the need for stronger policies to support those most affected.

Introduction and Background

The pandemic caused widespread economic instability, which worsened housing affordability, income inequality, and racial disparities. Los Angeles County, home to one of the most expensive housing markets in the United States, faced challenges as both renters and homeowners struggled under financial pressure. Rent burden became a major concern, especially for low-income and minority households, where many of them were affected by job losses and reduced work hours. For this study, we define rent burden as households spending 50% or more of their income on rent.

A significant driver of housing instability during the pandemic was income loss among minorities. Even before the pandemic, Latino and Black households were overrepresented in rent-burdened categories (Los Angeles County Economic Development Corporation, 2017). Data from the American Housing Survey (AHS, 2024) shows that over 50% of Latino renters in Los

Angeles spend more than 30% of their income on rent, categorizing them as rent-burdened. While federal relief programs, such as eviction moratoriums and direct stimulus payments, provided temporary assistance, they did not address the deeper crisis, raising concerns about long-term displacement.

Historically, economic downturns have increased the racial wealth gap, since lower-income households often lack financial safety nets. The 2008 housing crisis is an example, where Black and Latino homeowners faced high foreclosure rates due to predatory lending practices. A similar pattern emerged during the pandemic, especially since job losses were concentrated in the service sector, an industry where minority workers are overrepresented. As a result, census tracts where Latinos are the majority were particularly vulnerable to housing instability and economic shocks, continuing a trend seen in previous recessions (Los Angeles County Economic Development Corporation, 2017). At the same time, California's homeowners skew significantly white (CalMatters, 2023), reflecting long standing racial disparities in property ownership and wealth accumulation.

Beyond rent burden, homeownership trends also highlight racial and income disparities. The pandemic triggered an increase in real estate prices, further excluding lower-income households from homeownership. While high-income buyers capitalized on low-interest rates, middle and low-income renters faced greater obstacles, such as rising down payment requirements and heightened competition for affordable homes. Although homeownership rates increased slightly post-pandemic, this growth was primarily among high-income households, further deepening wealth inequality.

This study examines post-COVID shifts in rent burden, homeownership, and median income, focusing on disparities based on income and racial composition. By analyzing these trends, we aim to provide policymakers with data-driven insights into the long-term effects of COVID-19 on housing affordability, homeownership access, and economic mobility in Los Angeles.

Hypothesis

This study tests three key hypotheses about the impact of COVID-19 on housing outcomes in Los Angeles County.

Hypothesis 1: Rent burden increased after COVID-19, particularly for low-income and minority households. Since Black and Latino workers are overrepresented in service-sector jobs, they were more likely to experience job losses and wage stagnation. As rental prices rose while their incomes remained unchanged, their rent burden likely worsened. However, temporary relief measures, such as eviction moratoriums and stimulus payments, may have prevented extreme increases in rent burden (Deng, 2024).

Hypothesis 2: Homeownership rates declined post-COVID, especially among low-income households. Homeownership is a key driver of wealth accumulation, yet rising housing costs, wage stagnation, and restricted credit access have made it increasingly difficult for lower-income groups to buy homes. While historically low mortgage rates allowed high-income households to invest in property, middle and low-income renters faced growing affordability barriers, leading to further declines in homeownership.

Hypothesis 3: Median income increased for high-income households after COVID-19 but declined for lower-income groups, particularly in Black and Latino-majority census tracts. The economic recovery primarily benefited wealthier individuals, many of whom worked remotely and maintained financial stability, while lower-income workers faced job disruptions and stagnant wages. As a result, we expect income growth to be concentrated in predominantly White, high-income areas, while Black- and Latino-majority tracts experienced disproportionate declines (Mordechay et al., 2023).

By testing these hypotheses, this study aims to clarify how racial and economic disparities shaped post-pandemic housing trends in Los Angeles County. The findings will help inform policies to address housing insecurity and support a more equitable economic recovery.

Data and Methodology

This study takes a quantitative approach, using publicly available datasets to examine changes in rent burden, homeownership rates, and median income between the pre-pandemic (2014–2018) and post-pandemic (2018–2022) periods. The primary data source is the American Community Survey (ACS) 5-year estimates, which provide demographic and housing statistics at the census tract level.

The analysis focuses on three key dependent variables: rent burden percentage, which measures the share of household income spent on rent and serves as a primary indicator of affordability; homeownership rate, which reflects the percentage of occupied units that are owner-occupied and indicates long-term financial stability; and median income, which assesses economic disparities at the census tract level.

Independent variables include a post-COVID indicator, a binary variable distinguishing pre-pandemic (2014–2018) from post-pandemic (2018–2022) data; racial composition, which accounts for the proportion of White, Black, Latino, and Asian residents in each tract; and income groups, where households are classified as low, middle, or high income based on the 33rd and 66th percentile cutoffs of median income. Economic controls include the total number of renter-occupied units, the share of renters experiencing extreme rent burden, defined as spending over 50% of income on rent, and the homeownership rate, which functions as both a dependent and control variable to track ownership disparities.

To analyze these relationships, we use Ordinary Least Squares (OLS) regression with interaction terms and fixed effects at the census tract level. Fixed effects account for unobserved, time-invariant factors within tracts, while the inclusion of interaction terms isolates the impact of COVID-19 on housing affordability and economic conditions. Additionally, clustered standard errors at the tract level are applied to address spatial correlation in the data.

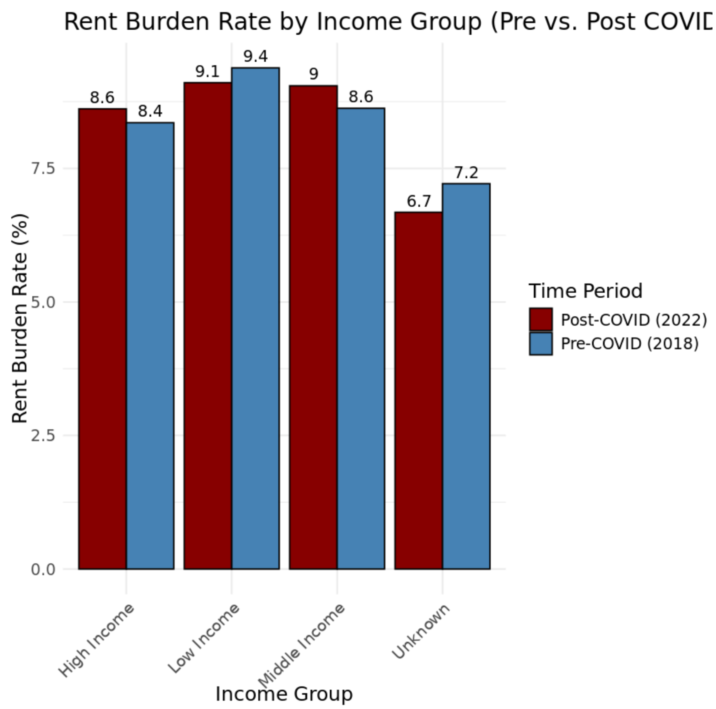
By implementing these econometric techniques, this study provides a comprehensive understanding of the structural inequalities exacerbated by the pandemic. The findings aim to inform policies that reduce long-term housing disparities and promote economic stability, particularly for vulnerable communities.

Results

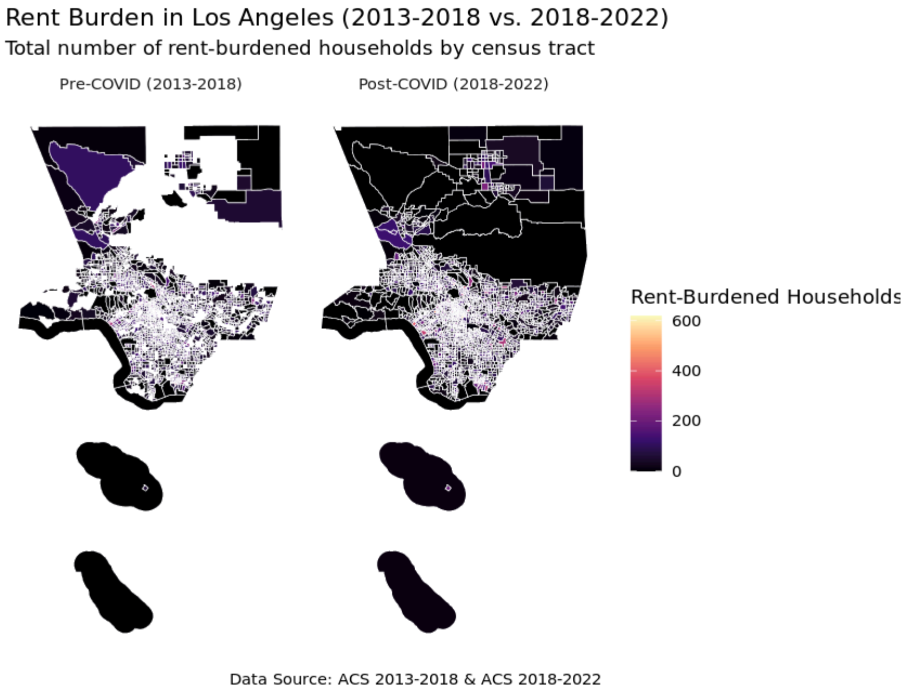
This analysis examines three key housing outcomes: rent burden, homeownership rates, and median income. Findings are supported by regression models that assess changes before and after COVID-19 while accounting for income levels, racial composition, and economic factors.

Rent Burden Analysis

Our findings reveal that rent burden remained relatively stable between 2018 and 2022, defying expectations of an increase after the pandemic. There were no statistically significant differences in rent burden rates across income groups during this period. This stability may be attributed to short-term federal relief programs, such as eviction moratoriums and rental assistance, which likely helped prevent a sharp rise in extreme rent burden. However, low-income renters continued to experience a higher rent burden compared to middle and high-income groups. While these relief measures provided temporary financial support, they did not resolve the deeper issue of unaffordable housing, which remains a severe issue in Latino and Black-majority census tracts.

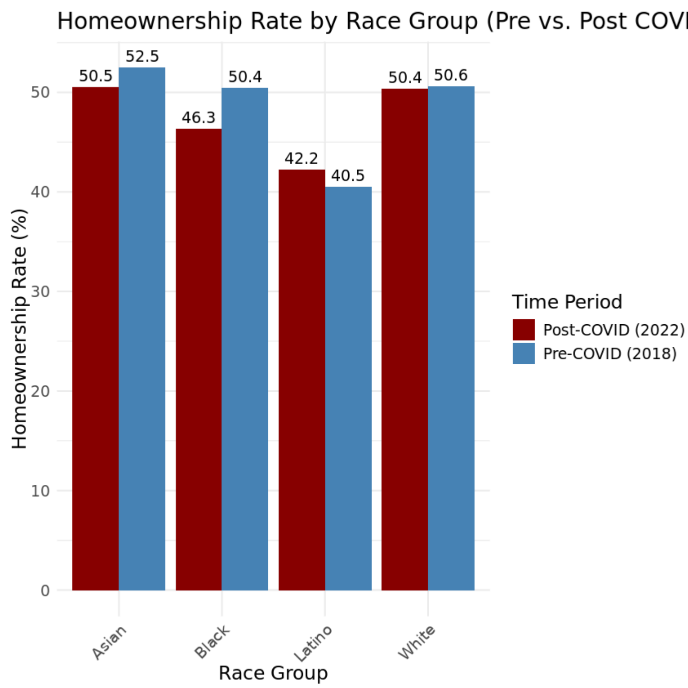


Smaller census tracts, which generally represent more densely populated urban areas, appear to be more affected by rent burden. The visualization shows that areas with a higher concentration of rent-burdened households tend to be in these densely populated urban regions, where housing demand is higher, affordability challenges are more severe, and rental costs are generally elevated. Larger census tracts, which usually represent less densely populated suburban or exurban areas, had little to no recorded rent burden pre-COVID but now show some increase post-COVID. However, this shift is likely due to new data availability rather than an actual affordability crisis, as remote work and migration trends brought more renters into these areas. Overall, the most persistent rent burden is still concentrated in historically affected urban areas, where lower-income and minority households are overrepresented, reinforcing long-standing affordability challenges despite short-term federal relief measures.

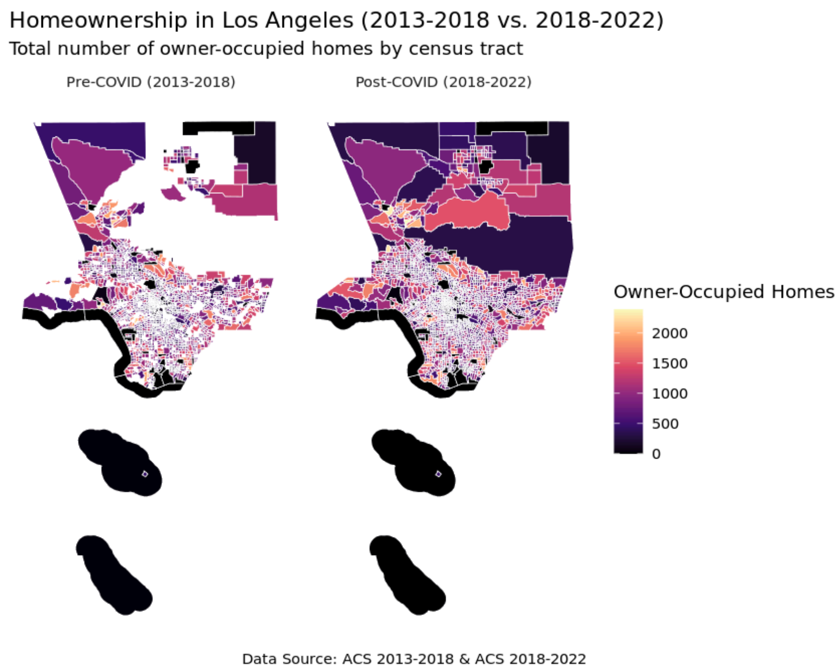


Homeownership Analysis

The chart shows a slight decline in Asian and White homeownership rates post-COVID, while Black homeownership saw the largest drop from 50.4% to 46.3%. In contrast, Latino homeownership increased slightly from 40.5% to 42.2%, though it remains the lowest among all groups. These trends suggest that rising housing costs and financial barriers impacted all racial groups to some extent, but Black households faced the greatest setbacks.

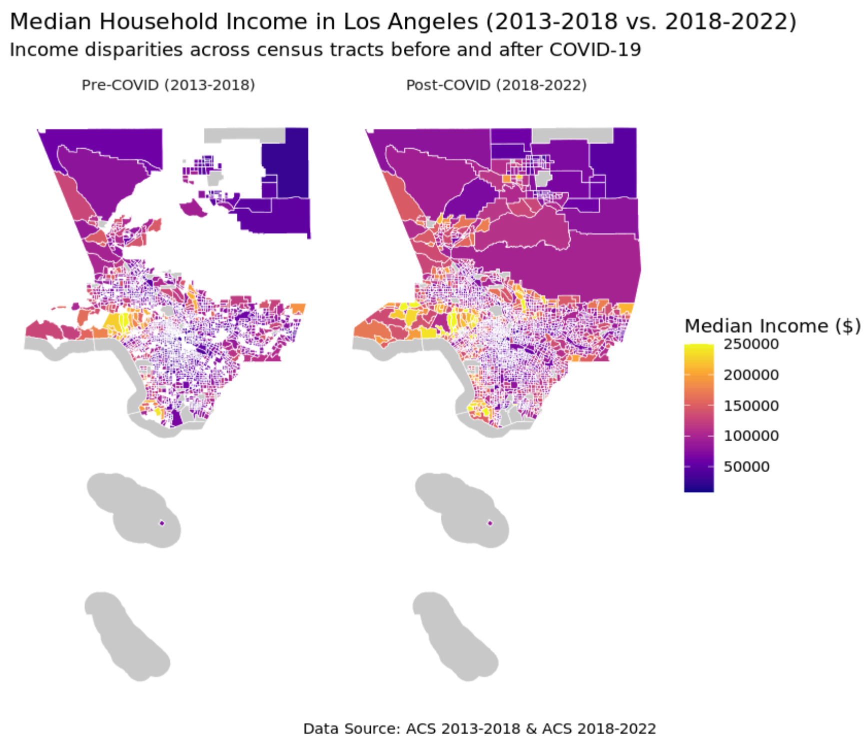


The post-COVID (2018–2022) map reveals an increase in owner-occupied homes in certain tracts, particularly in suburban and outer areas, suggesting that some households took advantage of historically low mortgage rates to transition from renting to homeownership. In contrast, high-density urban areas, especially in central Los Angeles, saw little change, likely due to high property costs and limited housing availability. Previously low-density areas now report higher homeownership rates, likely driven by remote work migration to more affordable locations. Despite these shifts, homeownership remains concentrated in specific neighborhoods, reinforcing pre-existing disparities. Many historically lower-income, renter-dominated areas continue to have low homeownership rates, highlighting the persistent affordability challenges that restrict access to homeownership for certain groups.

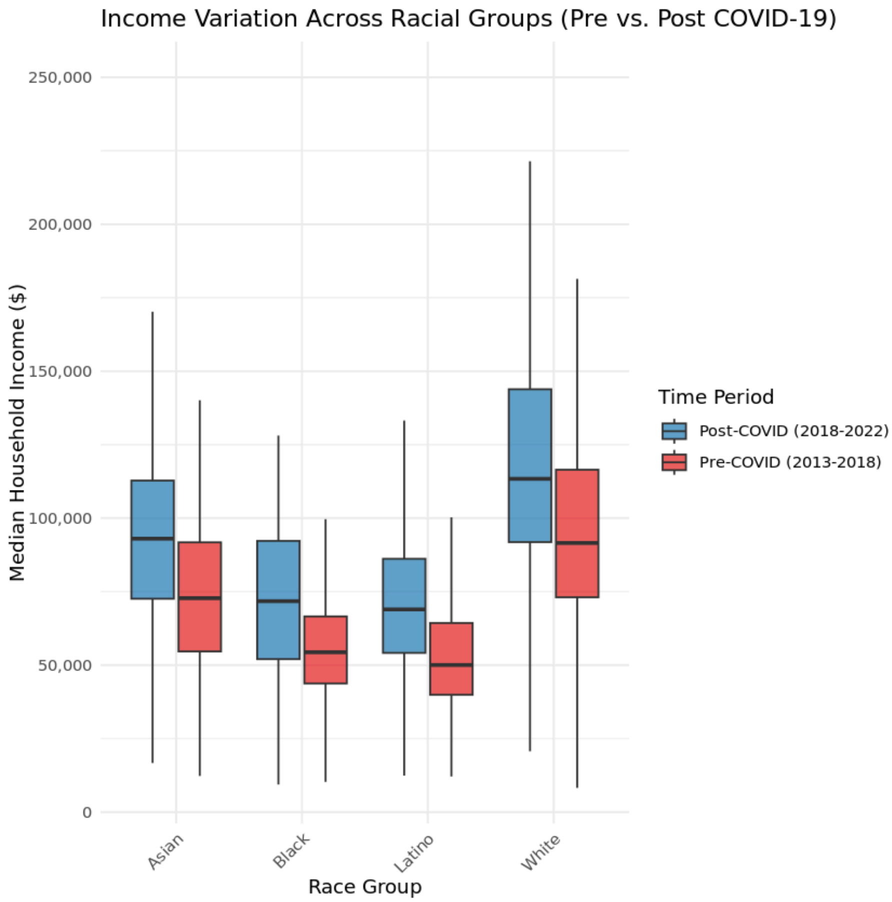


Median Income Analysis

The post-COVID (2018–2022) map shows a general increase in median income across many census tracts, particularly in wealthier areas, suggesting that some households benefited financially from stimulus checks, remote work opportunities, and growth in high-income sectors. However, income disparities widened, with western and coastal regions experiencing significant gains, while lower-income tracts in central, eastern, and southern Los Angeles saw little improvement or even declines. Wealthier neighborhoods, especially in the western and northern parts of the city, had the most substantial income growth, reinforcing pre-existing economic inequalities. Meanwhile, many lower-income communities, particularly in central and South Los Angeles, faced stagnant or declining median incomes, likely due to job losses in industries hardest hit by the pandemic, such as hospitality, retail, and service sectors. These patterns indicate that post-pandemic economic recovery has been uneven, disproportionately benefiting higher-income households while leaving many lower-income communities struggling.



White and Asian households had higher median incomes before and after COVID-19, but the post-pandemic data shows a growing gap between them and Black and Latino households, highlighting worsening economic inequality. While White and Asian households saw significant income gains, likely benefiting from economic recovery, remote work, and high-growth industries, Black and Latino households experienced only modest increases.



OLS Regressions

The first OLS regression evaluates rent burden changes between 2018 and 2022, controlling for homeownership rates, median income, total renters, extreme rent burden, and racial composition **(rent_burden_pct ~ post * income_group + homeownership_rate+ median_income + total_renters + extreme_rent_burden + white_pct + black_pct + latino_pct | GEOID)**. Results show a slight increase in rent burden post-COVID, though the effect is not statistically significant, suggesting that the pandemic had a limited impact on rent burden. Income group differences were also minimal, with middle-income households experiencing a slight decrease and high-income households a small increase, neither of which were statistically meaningful. The most significant predictors were total renters and extreme rent burden, indicating that areas with high pre-COVID rent burdens remained so, while increased rental supply may have helped stabilize costs. The model explains a substantial portion of rent burden variation, though the lack of significance in key predictors suggests that local policies and economic relief measures may have influenced post-pandemic trends.

The second regression examines changes in homeownership rates between 2018 and 2022, controlling for median income, total renters, extreme rent burden, and racial composition **(homeownership_rate ~ post * income_group + rent_burden_pct + median_income + total_renters + extreme_rent_burden + white_pct + black_pct + latino_pct | GEOID)**. Results indicate that homeownership increased post-COVID, High-income households saw a surprising drop in homeownership post-COVID, potentially reflecting shifts in real estate investments, affordability issues, or migration trends . Middle-income households saw little change, while homeownership rates among high-income households declined when accounting for interaction effects. Among control variables, higher median income was positively associated with homeownership, while a larger renter population reduced homeownership rates. Latino-majority areas experienced a notable decline in homeownership, whereas White and Black population proportions were not significant predictors. With an adjusted R^2 of 0.987, the model accounts for much of the variation in homeownership, though factors such as housing policies and lending practices likely influenced post-pandemic patterns.

The third regression explores how homeownership rates were influenced by post-COVID trends, racial composition, and income levels while controlling for census tract fixed effects (**homeownership_rate ~ post * (black_pct + latino_pct + asian_pct) +median_income + total_renters + extreme_rent_burden + income_group | GEOID**). Results indicate that Latino-majority tracts had significantly lower homeownership rates, while Black and Asian population proportions were not significant predictors. The post-COVID period did not substantially alter homeownership trends across racial groups. Income was a major factor, with high-income individuals significantly more likely to own homes, while middle-income status showed no meaningful effect. Low-income tracts continued to have the lowest homeownership rates. Higher median income was positively associated with homeownership, while areas with more renters had significantly lower homeownership rates. However, extreme rent burden was not a significant predictor, reinforcing that income remains the strongest determinant of homeownership, while race and pandemic-related shifts had limited effects.

The fourth regression assesses how rent burden was influenced by post-COVID trends, racial composition, and economic factors while controlling for census tract fixed effects **(rent_burden_pct ~ post * (black_pct + latino_pct + asian_pct) +median_income + total_renters + extreme_rent_burden + income_group | GEOID**). Results show that racial composition did not significantly predict rent burden, and there were no major post-COVID changes across racial groups. Higher median income was slightly associated with lower rent burden, though not significantly. Income group effects were also minimal, as middle and high-income groups did not experience significantly different rent burdens compared to low-income groups. Among economic factors, areas with more renters tended to have lower rent burdens, while extreme rent burden was strongly correlated with overall rent burden. The post-COVID period did not have a significant impact on overall rent burden, suggesting broader economic conditions and housing policies may have played a stabilizing role.

The fifth regression examines how median income was affected by post-COVID trends, racial composition, and housing factors while controlling for census tract fixed effects (**median_income ~ post * (black_pct + latino_pct + asian_pct) +homeownership_rate + total_renters + extreme_rent_burden + income_group | GEOID**). Results show that median income increased significantly post-COVID (+$26,414), but this growth was unevenly distributed. Before COVID, Latino and Asian-majority areas had higher median incomes, but post-COVID, these groups, along with Black-majority tracts, experienced declines. Latino-majority areas saw the most significant drop (-$117.56 per percentage point), followed by Black-majority areas (-$75.98 per percentage point), indicating that these communities faced disproportionate economic setbacks. Higher homeownership rates were linked to higher median income, while areas with more renters had lower median incomes. Extreme rent burden did not significantly impact median income. Middle and high-income tracts had significantly higher median incomes than low-income tracts, with high-income areas experiencing the largest post-COVID income gains. These findings highlight how overall income levels rose after the pandemic, yet racial and economic disparities in income distribution widened, particularly for historically marginalized communities.

Discussion and Policy implication

Rent Burden Analysis

The study highlights that rent burden remains relatively stable with no statistically significant difference across income groups. The stability may be attributed to short-term federal relief programs, such as eviction moratorium and rental assistance, which likely helped prevent sharp increase in rent burden. However, low-income renters continue to experience significantly higher rent burden than middle and high income groups. This finding aligns with statewide data showing that low income households face the highest cost pressures, with around 2.5 million low income households in California being cost burdened (Legislative Analyst's Office [LAO], 2023). In addition to that, some rural areas with little to no rent burden data pre-COVID (2014-2018) now report rent-burdened households post-COVID (2018-2022). This change may not be due to actual increase in rent burden but availability of new data, as remote work enables more people to move to previously low-density areas (Ray et al., 2021). These areas previously have fewer renters or are sparsely populated which limits rent burden data . With increased migration to more affordable locations, those areas now reflect housing affordability challenges that were previously unrecorded. This shift underscores how remote work reshaped housing patterns, extending rental pressures beyond traditional high-cost urban centers (Angst et al., 2021).

Homeownership Analysis

The effect of Covid-19 on homeownership reveals a divide across different income groups. While the OLS regression shows that homeownership rates increase slightly overall, it tends to decline for higher income groups post-Covid 19. White majority census tracts saw modest increases and Latino majority continued to have the lowest homeownership.This pattern reflects a border economic trend where white benefit from low mortgage interest rate and expanded homeownership, latino majority faced rising home prices and financial barriers(LAO, 2023). Additionally, post pandemic job instability among lower-income groups limited their ability to transition into homeownership further reinforcing wealth disparity (Ray et al., 2021). Moreover, some previously low-density areas now report higher homeownership rates, likely driven by remote work migration to more affordable locations. Despite this shift, homeownership remains concentrated in specific neighborhoods, reinforcing pre-existing disparities (Mordechay et al., 2023).

Median Income Analysis

The post-COVID (2018-2022) map illustrates a general increase in median income across many census tracts, particularly in wealthier areas, suggesting that some households benefit financially from stimulus checks, remote work opportunities, and growth in high-income sectors (Ray et al., 2021). However, visualization also illustrates that the income disparities widen with western and coastal regions experiencing significant gains, low income tracts in central, easton and southern LA saw little improvement or even declines. Wealthier neighborhoods had the most substantial income growth indicating pre-existing economic inequalities (Mordechay et al., 2023). At the same time, many lower income communities face declining median incomes, likely due to job losses in industries hardest hit by the pandemic, such as hospitality, retail and service sector (Angst et al., 2021).

White and Asian households maintained higher median income both before and after COVID-19, but post-pandemic data shows a widening gap between these groups and black and latino households, indicating worsening economic stratification (Mordechay et al., 2023). This trend shows long standing income inequality as post pandemic economic growth disproportionately benefited already in higher income brackets (Ray et al., 2021).

Implication

This study highlights persistent challenges in housing affordability, homeownership, and income inequality in Los Angeles, exacerbated by the COVID-19 pandemic. While federal and state aid provided temporary relief, deeper structural issues remain. To promote fair housing opportunities, policies must focus on expanding affordable housing, increasing homeownership accessibility, addressing income disparities, and preventing displacement.

Despite stable rent costs, low-income and minority renters—particularly in Latino- and Black-majority neighborhoods—continue to face high rent burdens. Expanding affordable housing programs through tax credits and subsidies can incentivize developers to build more affordable units. Strengthening rent control laws, expanding Section 8 vouchers, and establishing permanent eviction protections for those facing financial hardship can help stabilize rental costs and prevent displacement.

Homeownership rates increased slightly post-pandemic, but gains were concentrated among high-income households, leaving Latino- and Black-majority neighborhoods with persistently lower homeownership rates. Expanding first-time homebuyer programs, offering low-interest mortgage loans, and providing down payment assistance for low- and middle-income families can help address these disparities. Strengthening protections against housing discrimination and unfair lending practices is also essential.

While median income rose overall, racial income gaps widened, with White-majority neighborhoods experiencing significant growth while Latino- and Black-majority areas saw declines. To ensure a more equitable economic recovery, policies should focus on expanding job training and workforce development, raising minimum wages, and enforcing pay transparency. Direct cash assistance, tax relief for low-income families, and investment in minority-owned businesses can help stimulate local economies and create jobs in disadvantaged communities.

Geographic analysis shows that historically marginalized neighborhoods continue to face the greatest challenges. To prevent displacement and ensure long-term housing stability, expanding community land trusts (CLTs) can help protect affordable housing from gentrification. Reforming zoning laws to encourage mixed-income housing without displacing residents and investing in public transportation and job access in high-rent-burden areas can create economic opportunities. Strengthening rent stabilization programs in rapidly gentrifying neighborhoods can also protect low-income and minority residents from displacement.

While temporary COVID-19 relief efforts helped stabilize rent burdens, long-standing issues in homeownership, income inequality, and housing affordability persist. Without targeted policies, these disparities will continue, keeping marginalized communities at an economic disadvantage. Expanding affordable housing, improving homeownership accessibility, reducing income inequality, and preventing displacement are critical steps toward a more just and inclusive housing system where all Los Angeles County Residents—regardless of income or race—have access to stable, affordable housing and economic opportunities.

Conclusion

This study brings to light the inequalities in housing affordability, homeownership, and income distribution in Los Angeles County and its relation to the COVID-19 pandemic. Temporary relief programs, such as eviction moratoriums and rental assistance, helped prevent a housing crisis but they did not address the long-standing economic and racial disparities shaping housing access.

Between 2018 and 2022, rent burden remained stable, suggesting that short-term government programs helped prevent extreme increases. However, low-income and minority households continue to struggle with high rent costs, highlighting ongoing affordability issues. Homeownership rates rose post-pandemic, but gains were concentrated among high-income households, while Latino- and Black-majority neighborhoods saw little improvement, reinforcing barriers to property ownership and wealth-building.

Racial and economic income gaps also widened. While median income increased overall, White-majority areas saw the most significant growth, while Latino- and Black-majority neighborhoods experienced declines, further deepening wealth inequality. Spatial analysis confirms that historically marginalized communities in South and East Los Angeles continue to face the highest rent burden and lowest homeownership rates, while wealthier, predominantly White neighborhoods experienced the largest income gains.

These findings underscore the need for long-term policies to improve housing accessibility and economic equity. Expanding affordable housing, increasing homeownership opportunities for low- and middle-income households, addressing racial income disparities, and preventing displacement in historically marginalized areas are critical steps toward a more equitable housing system.

As Los Angeles moves toward economic recovery, prioritizing equity in housing policies is essential. Without action, racial and economic disparities in rent burden, homeownership, and income will continue to grow. By implementing evidence-based solutions—such as incentives for affordable housing, homeownership assistance, wage protections, and anti-displacement measures—policymakers can create a fairer housing system where all residents, regardless of income or background, have access to stable and affordable housing, promoting long-term economic mobility and social equity.

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